Geographical distribution of the 25 candidate cities

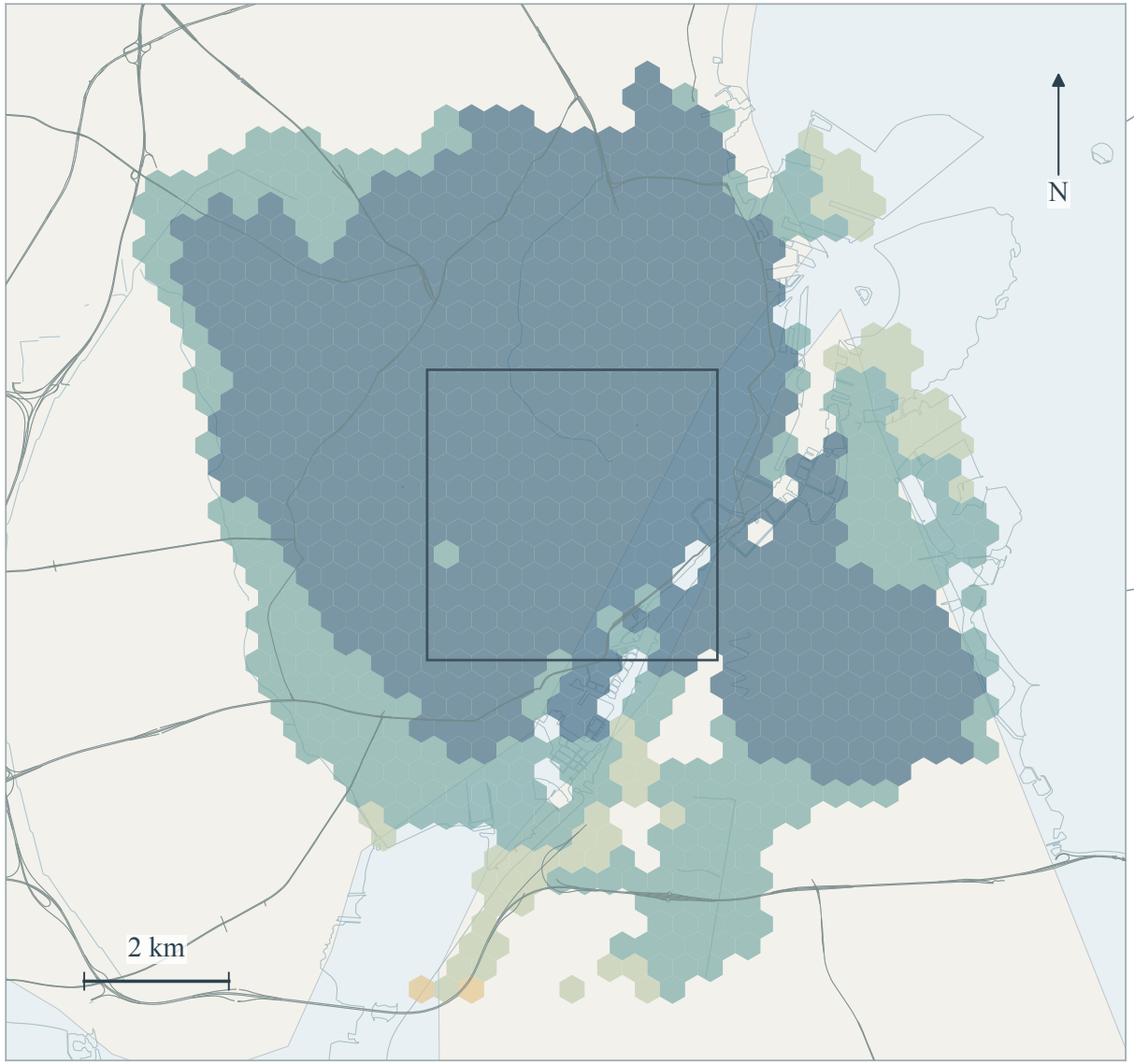
Study locations from the archived city index

25 cities · all candidates shown · Europe enlarged for legibility

Made with Natural Earth | Robinson projection | Urban-area reference locations

Copenhagen

Learning-facility accessibility across retained populated grids



Selected neighbourhood detail

Within 30 min: 99.4% population
Over 60 min: 0.0% population

Modelled walking time (min)

- 0–15
- >15–30
- >30–60
- >60–90
- >90

Selected OSM roads

Mean walking time to 20 nearby learning POIs · Population-weighted summaries
Five time classes · Urban population grids · Independent inset scale

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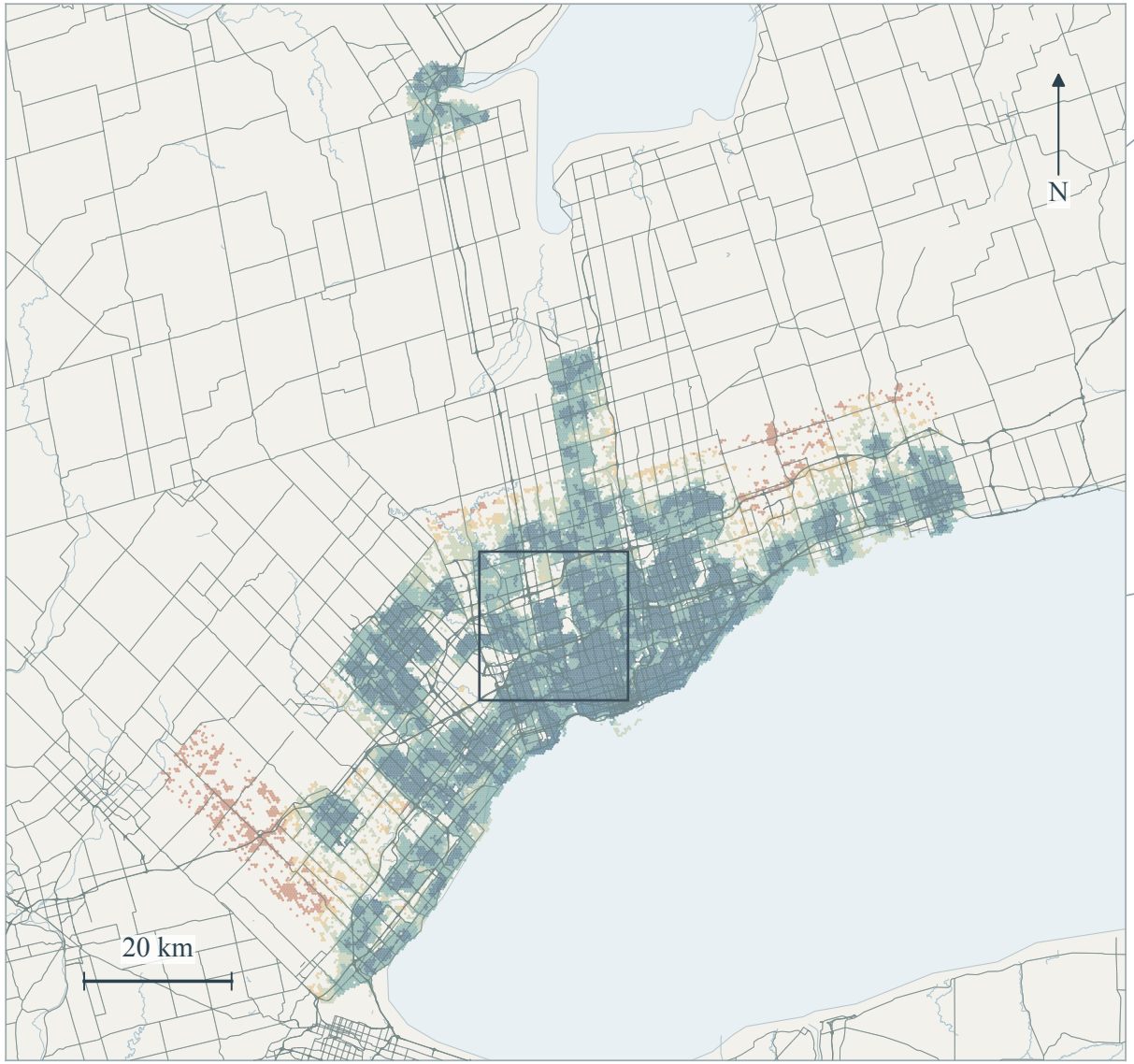
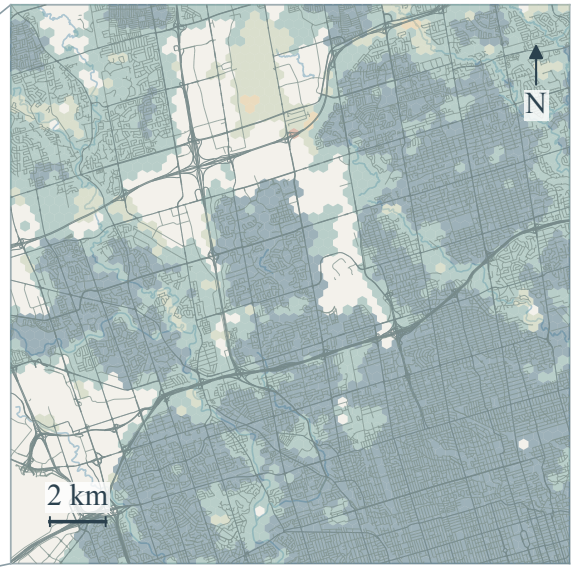
Toronto

Learning-facility accessibility across retained populated grids



Selected neighbourhood detail



Within 30 min: 98.2% population
Over 60 min: 0.3% population

Modelled walking time (min)

- 0–15
- >15–30
- >30–60
- >60–90
- >90

Selected OSM roads

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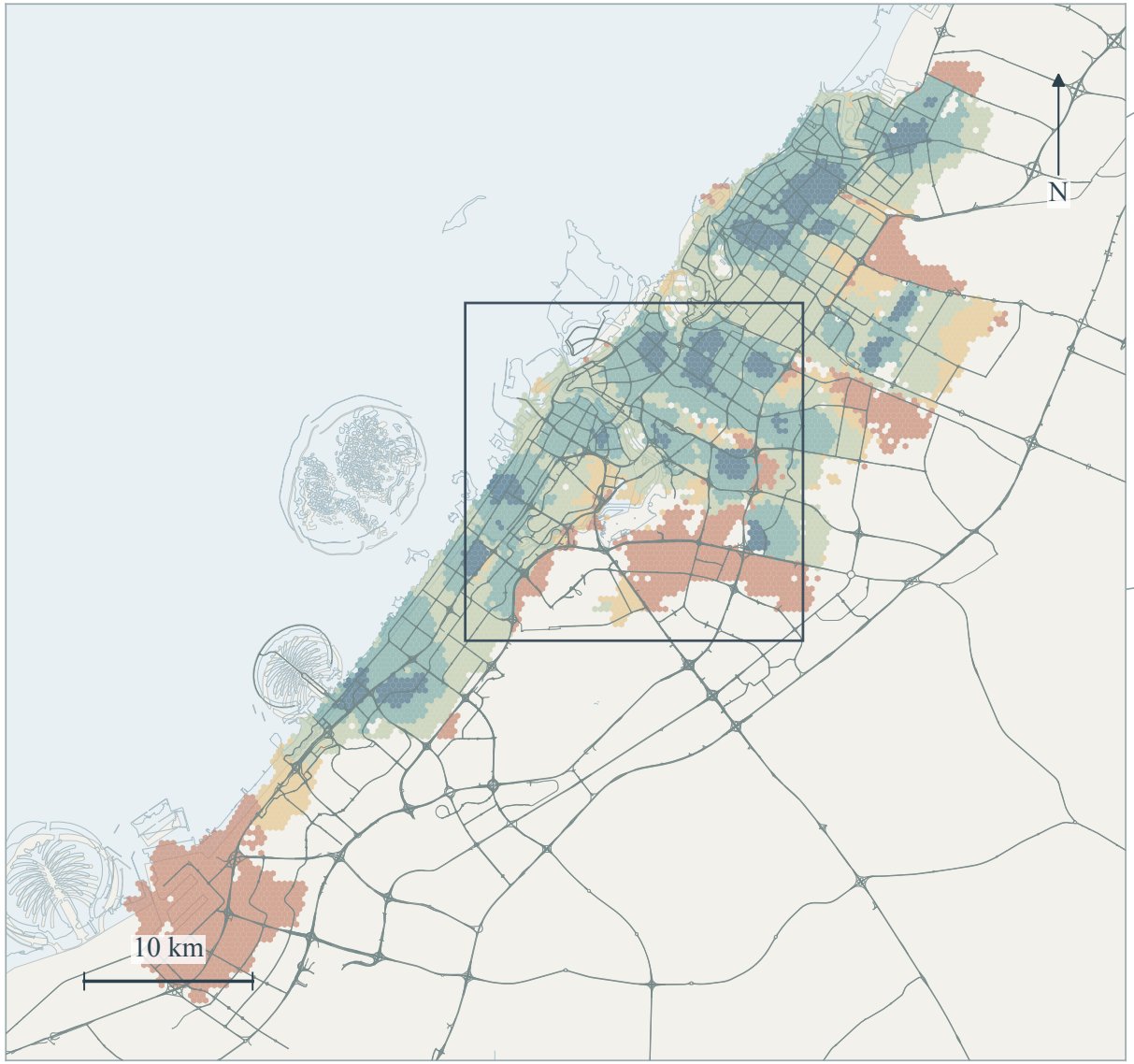
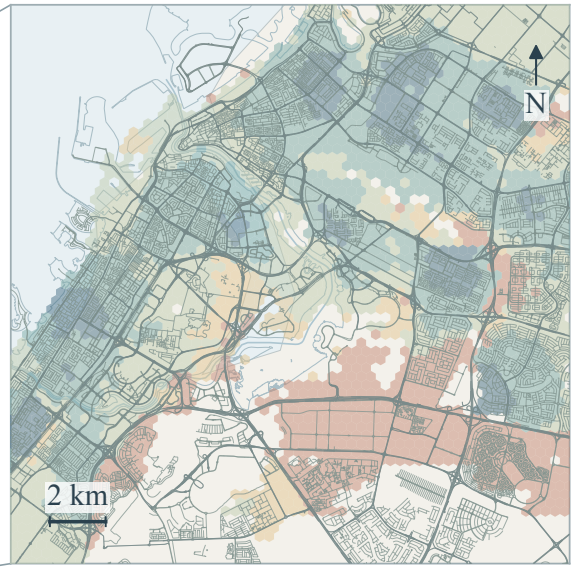
Dubai

Learning-facility accessibility across retained populated grids



Selected neighbourhood detail



Within 30 min: 43.7% population
Over 60 min: 23.4% population

Modelled walking time (min)

- 0–15
- >15–30
- >30–60
- >60–90
- >90

Selected OSM roads

Mean walking time to 20 nearby learning POIs · Population-weighted summaries
Five time classes · Urban population grids · Independent inset scale

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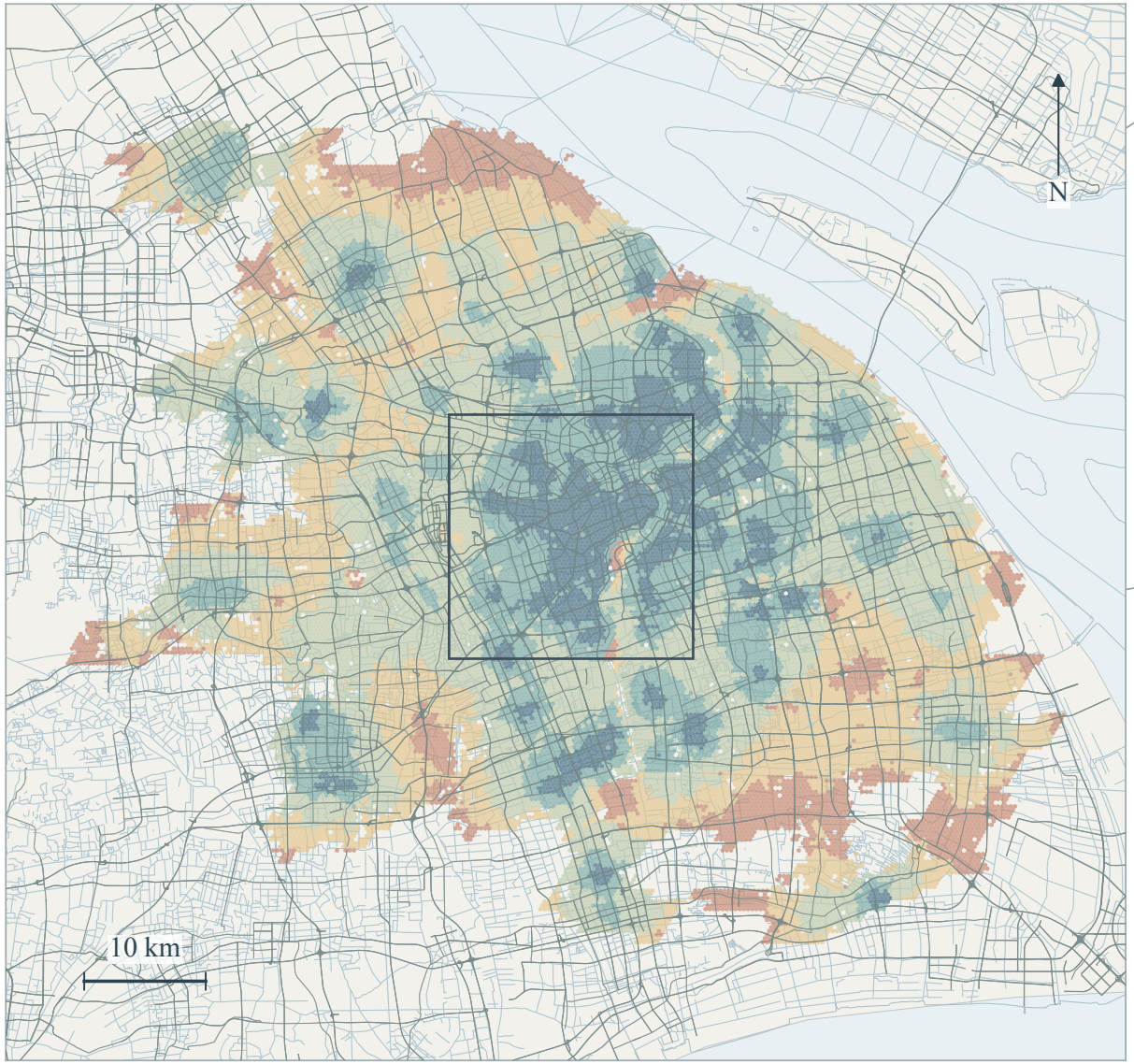
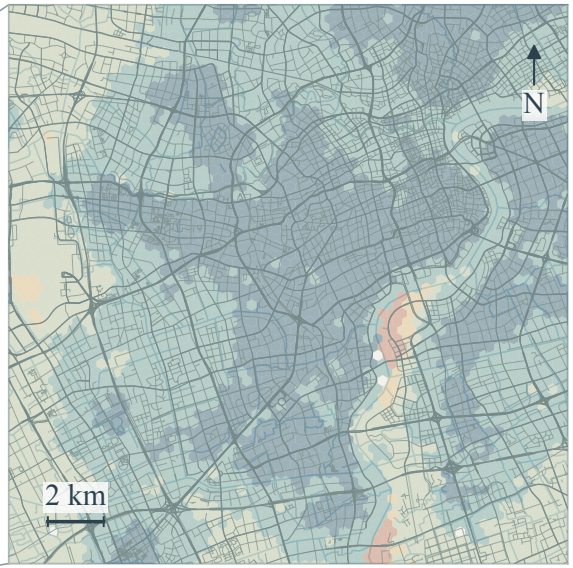
Shanghai

Learning-facility accessibility across retained populated grids



Selected neighbourhood detail



Within 30 min: 58.2% population
Over 60 min: 13.4% population

Modelled walking time (min)

- 0–15
- >15–30
- >30–60
- >60–90
- >90

Selected OSM roads

Mean walking time to 20 nearby learning POIs · Population-weighted summaries
Five time classes · Urban population grids · Independent inset scale

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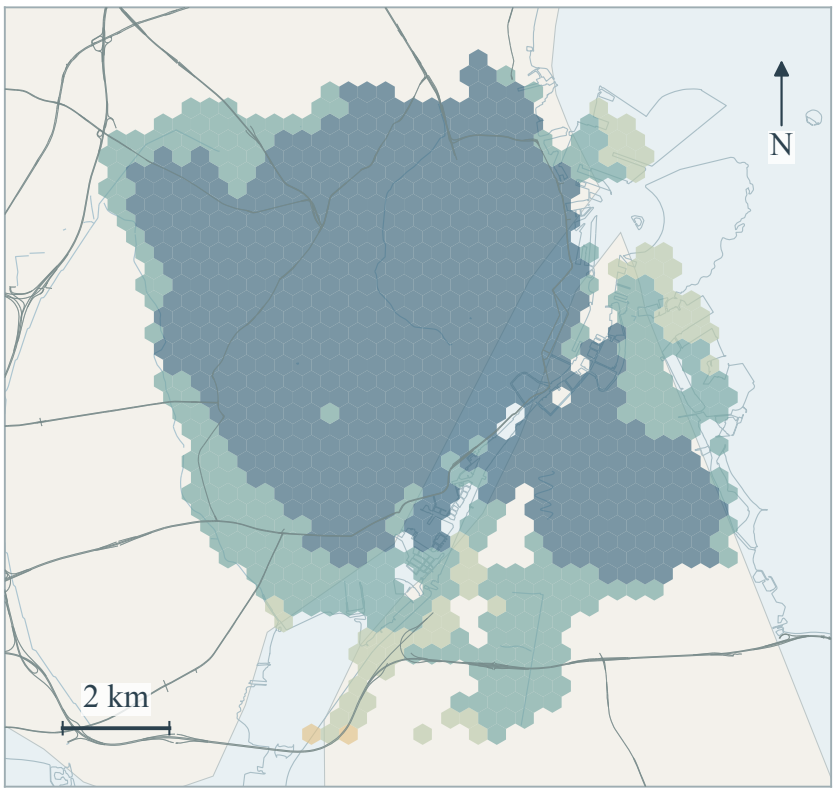
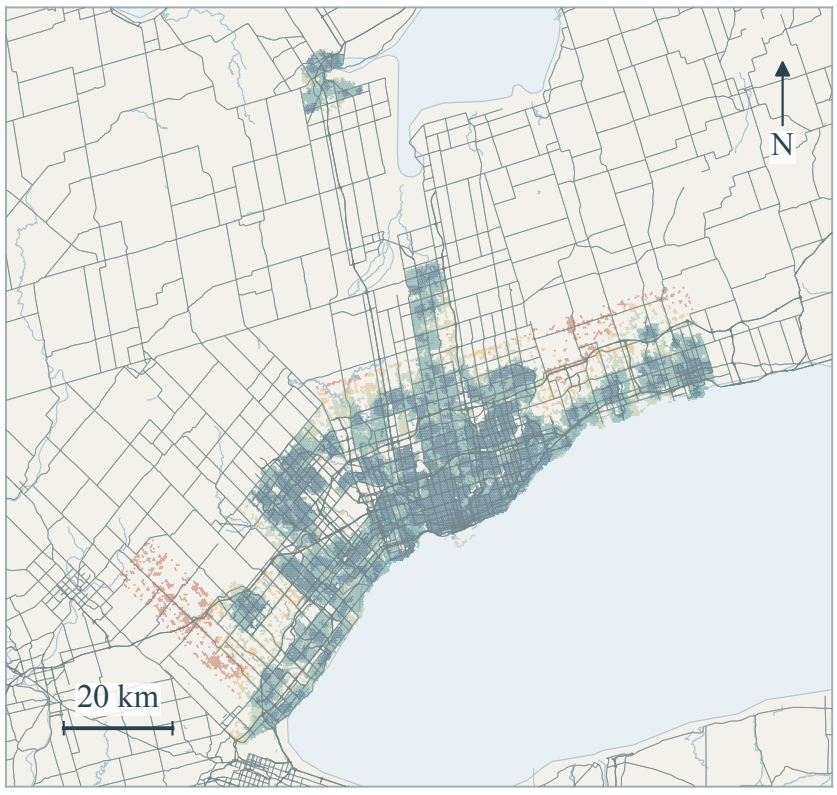
City-scale patterns of learning-facility access

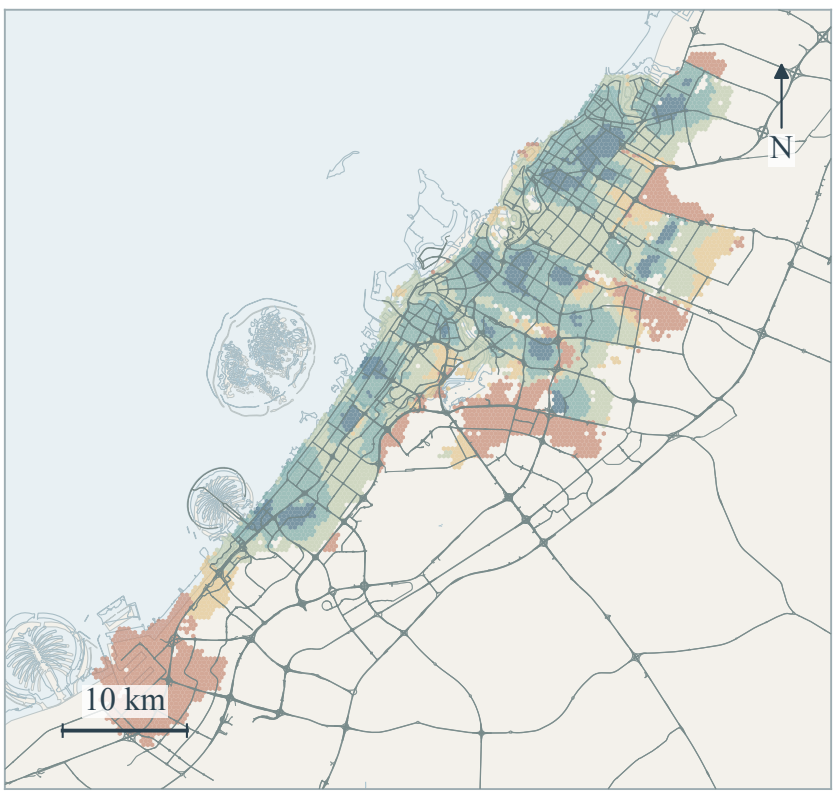
(a) Copenhagen



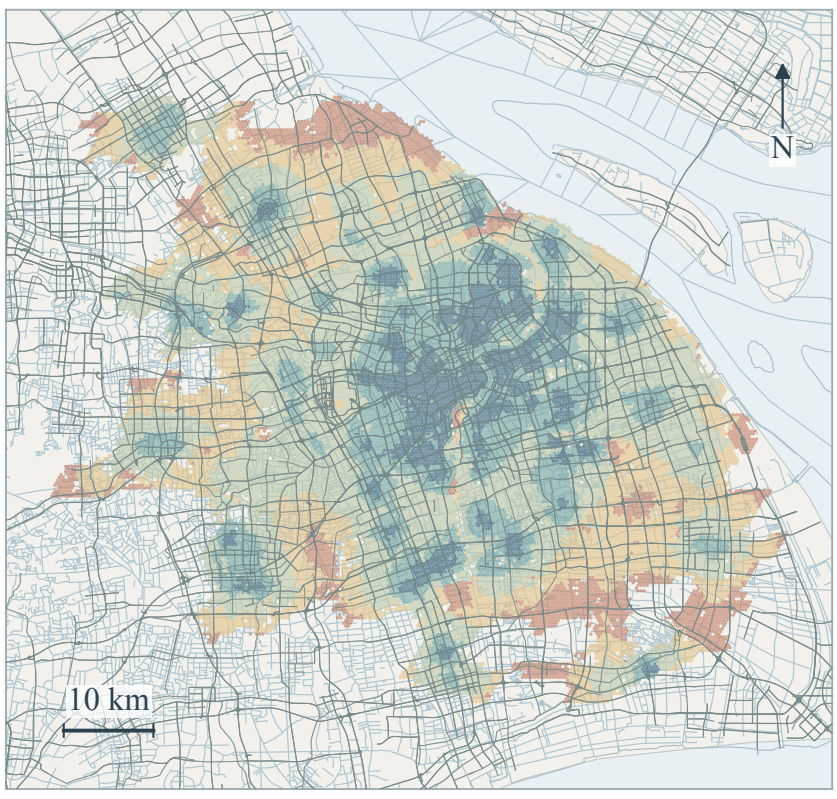
(b) Toronto



(c) Dubai



(d) Shanghai



Modelled walking time (min)

0–15 >15–30 >30–60 >60–90 >90

Shared time classes · City-specific scales · Populated urban grids

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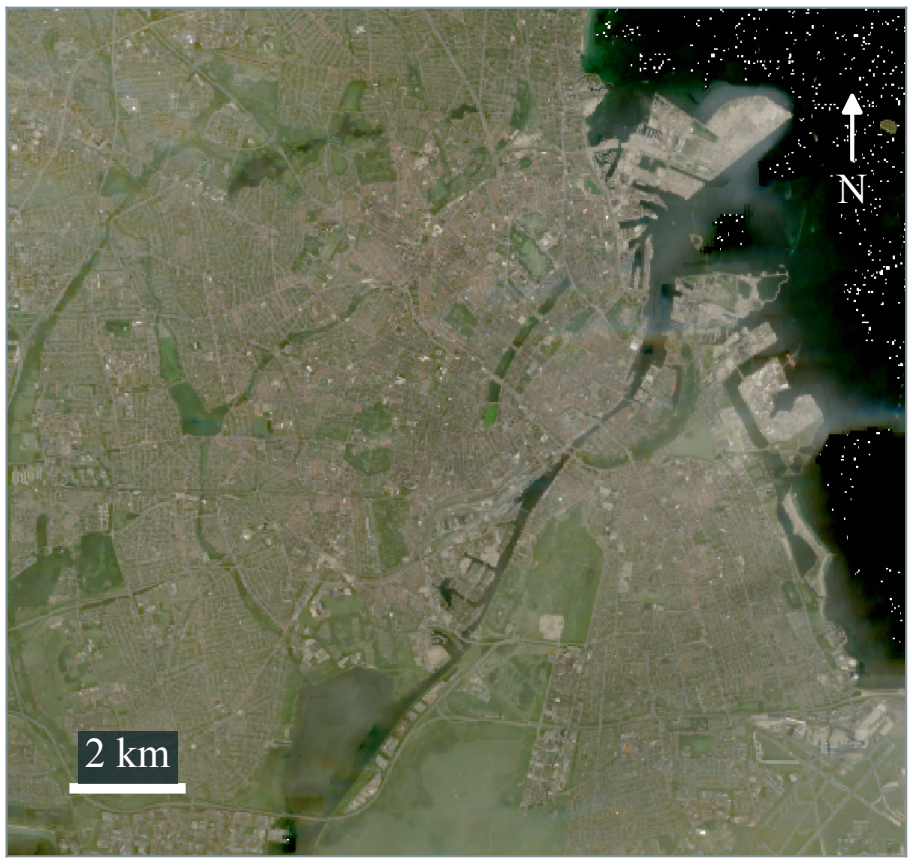
Satellite context for city accessibility

Sentinel-2 / HLS S30 · matched map extents · dated observations

(a) Copenhagen | 2024-06-27



(b) Toronto | 2024-10-17

(c) Dubai | 2024-09-04

(d) Shanghai | 2025-04-07

NASA HLS / GIBS · 30 m source product · EPSG:3857 · city-specific scales